

EXAM 7 – SPRING 2012

14. (2 points)

The Chief Risk Officer for an insurance company is considering U.S. Treasury interest rate models for the company's Enterprise Risk Management process.

a. (0.75 point)

Provide the formula for the Vasicek model and briefly describe the process it models.

b. (0.5 point)

Describe the key difference between the Cox, Ingersoll, Ross (CIR) model and the Vasicek model.

c. (0.75 point)

Provide and support a recommendation as to which of the two models, Vasicek or CIR, is more appropriate to use in the current environment of historically low interest rates.

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Question 14 Sample Answer

Solution 1

- a) $dr_t = (b - ar_t)dt + sdw_t \rightarrow$ form of Vasicek model
- Models a mean reverting Brownian motion where b is the mean term and a corresponds to the autocorrelation
- b) CIR model is $dr_t = (b - ar_t)dt + s\sqrt{r} dw_t$
- It introduces the rate as a factor of the variation which allows the variation to vary with the rate.
- c) I would recommend the CIR model because it has the added feature where the square root of the rate is used in the variability. This means that the volatility is linked to the rate, so in the current environment in low interest rates there is limited or little volatility. Also because of this feature, rates can never be negative, which in the current environment of low interest rates is a good thing because we could potentially model negative rates otherwise.

Solution 2

- a) $dr_t = (b - ar_t)dt + sdW_t$
- Vasicek is mean reverting Brownian motion model for the short rate r , with mean b/a
- b) CIR model: $dr_t = (b - ar_t)dt + s\sqrt{r} dW_t$
- The key difference is the inclusion of the $\sqrt{r}$ term in the standard deviation. This term makes it so the CIR model can't produce negative rates and so that volatility increases with the rate, which is seen empirically.
- c) CIR is more appropriate as it is a more robust model. Because interest rates are so low, there is an increased risk of Vasicek producing a negative rate, which is impossible. CIR doesn't have this issue. Also, it is seen empirically that volatility does not increase with higher rates, so this feature of the CIR model is preferable to include.

Examiner Comment

For the a. part, in addition to the formula, a brief description of the Vasicek model using at least 2 key words such as “mean reverting”, “Brownian motion”, “short rate”, “continuous” was required to receive full credit. It was not sufficient to only say that the Vasicek Model models US treasury interest rates since this is defined in the problem.

Some candidates used alternative letters for r , a , b , etc. when defining the formula. Full credit was provided as long as the formula was correct.

Common errors included:

- Calling the Vasicek model a discrete distribution rather than a continuous distribution
- Not including a dt in the first term (the dt shows that this is a continuous distribution, which is a key point in the model)
- Using r in the second term
- Using $a(b-r_t)dt$ as the first term, as the b term is not multiplied by a
- Saying that the model measured interest rates, as this was defined in the problem
- Using just 1 key word in the description

For the b. part, the key difference is the addition of the square root of r term in the formula which does not allow the CIR model to produce negative rates. Partial credit was given for a number of alternative differences between the CIR model and the Vasicek model, as long as the difference was accurate.

One common error was that candidates took the square root of s rather than r .

For the c. part, candidates must include discussion about the low interest rate environment when selecting the appropriate model. Only partial credit was awarded if the low interest rate environment was not discussed.